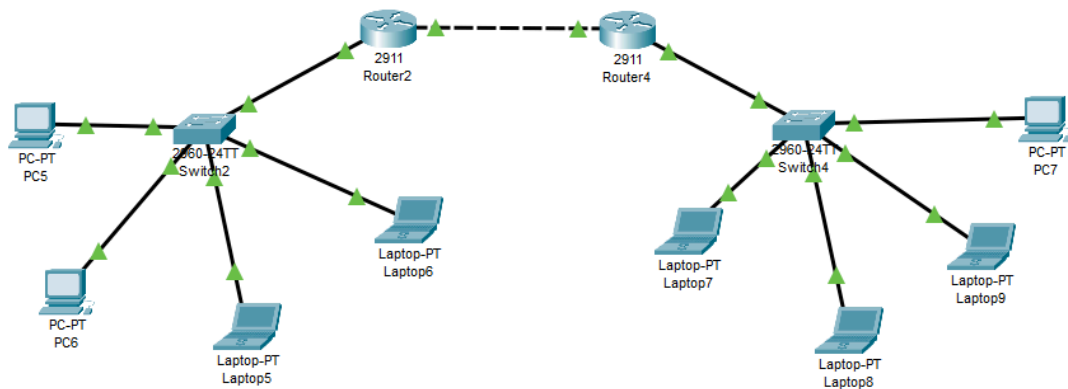


SMALL OFFICE MULTISITE NETWORK

This is the main project structure.

PC5, PC6, Laptop5, Laptop6 -> Switch2 -> Router2 <-> Router4 <- Switch4 <- PC6, Laptop7, Laptop8, Laptop9



I implemented a multi-site VLAN network with DHCP configuration, which was moderately challenging but highly insightful.

First, I configured VLANs on the switches and established trunk links between the switches and routers. Then, I implemented inter-VLAN routing using the router-on-a-stick method. DHCP services were configured directly on the routers to dynamically assign IP addresses to client devices in different VLANs.

DHCP Configuration:

I created separate DHCP pools on each router for the respective VLANs to ensure proper IP address allocation for all departments.

Router Configuration (R1 – Office A):

The router was configured as follows:

```
enable
```

```
configure terminal
```

```
interface g0/0
```

```
no shutdown
```

```
interface g0/0.10
```

```
encapsulation dot1Q 10
```

```
ip address 192.168.10.1 255.255.255.0
```

```
interface g0/0.20
```

```
encapsulation dot1Q 20
```

```
ip address 192.168.20.1 255.255.255.0
```

```
interface g0/1
```

```
ip address 10.0.0.1 255.255.255.252
```

```
no shutdown
```

```
ip dhcp excluded-address 192.168.10.1
```

```
ip dhcp excluded-address 192.168.20.1
```

```
ip dhcp pool HR
```

```
network 192.168.10.0 255.255.255.0
```

```
default-router 192.168.10.1
```

```
dns-server 8.8.8.8
```

```
ip dhcp pool IT
```

```
network 192.168.20.0 255.255.255.0
```

```
default-router 192.168.20.1
```

```
dns-server 8.8.8.8
```

```
ip route 192.168.30.0 255.255.255.0 10.0.0.2
```

```
ip route 192.168.40.0 255.255.255.0 10.0.0.2
```

Router Configuration (R2 – Office B):

enable

configure terminal

interface g0/0

no shutdown

interface g0/0.30

encapsulation dot1Q 30

ip address 192.168.30.1 255.255.255.0

interface g0/0.40

encapsulation dot1Q 40

ip address 192.168.40.1 255.255.255.0

interface g0/1

ip address 10.0.0.2 255.255.255.252

no shutdown

ip dhcp excluded-address 192.168.30.1

ip dhcp excluded-address 192.168.40.1

ip dhcp pool SALES

network 192.168.30.0 255.255.255.0

default-router 192.168.30.1

dns-server 8.8.8.8

ip dhcp pool FINANCE

network 192.168.40.0 255.255.255.0

default-router 192.168.40.1

dns-server 8.8.8.8

ip route 192.168.10.0 255.255.255.0 10.0.0.1

ip route 192.168.20.0 255.255.255.0 10.0.0.1

Key Concept:

Inter-VLAN routing was achieved using subinterfaces on a single physical interface (g0/0). Each subinterface was assigned to a specific VLAN using encapsulation dot1Q. The WAN link between routers was configured on a separate physical interface (g0/1).

Final Setup:

After completing the router and switch configurations, I connected the PCs to their respective VLANs. The switches were configured with access ports and a trunk link to the router.

All PCs were set to obtain IP addresses dynamically using DHCP. The devices successfully received IP addresses from their respective VLAN pools, confirming correct DHCP configuration.

Additionally, successful ping tests between devices in different VLANs and across both offices verified that inter-VLAN routing and inter-network communication were functioning correctly.

-> Sk. Md. Rashid Assef Shibly